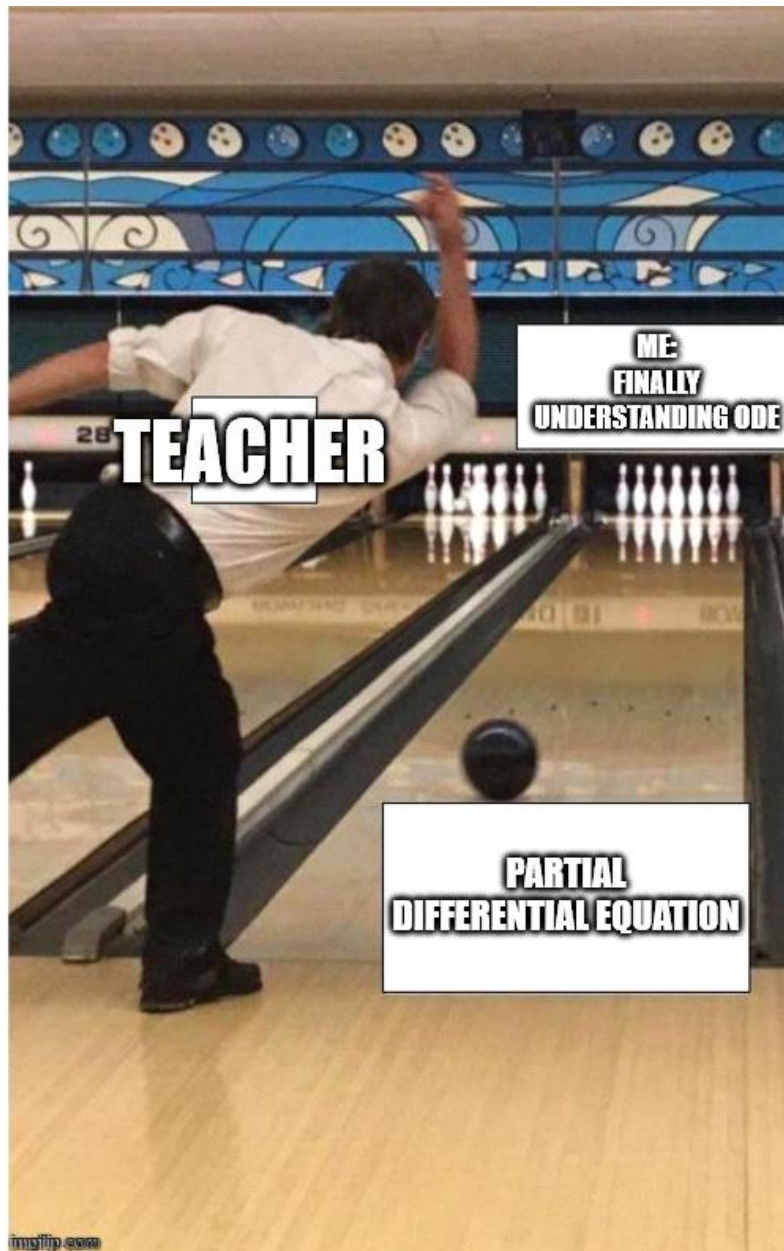
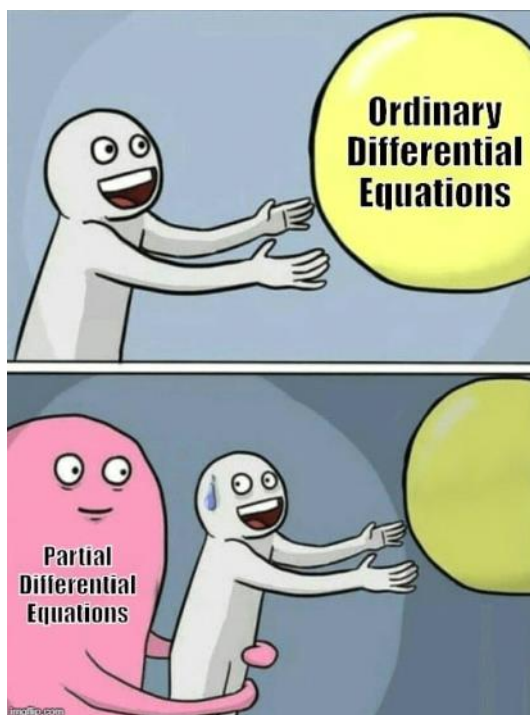
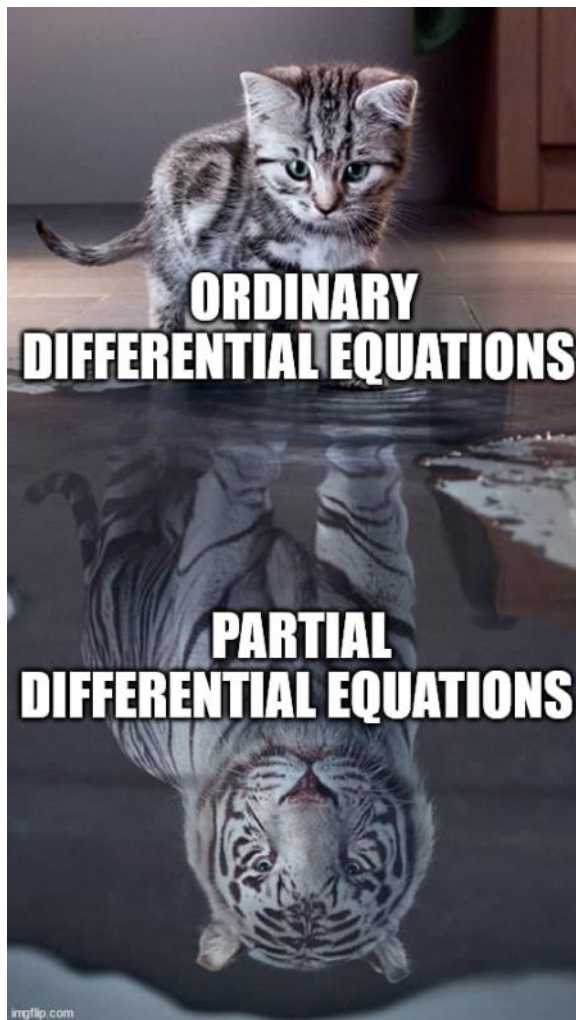


Partial Differential Equation

Tuesday, 5 May 2026

12:06 pm

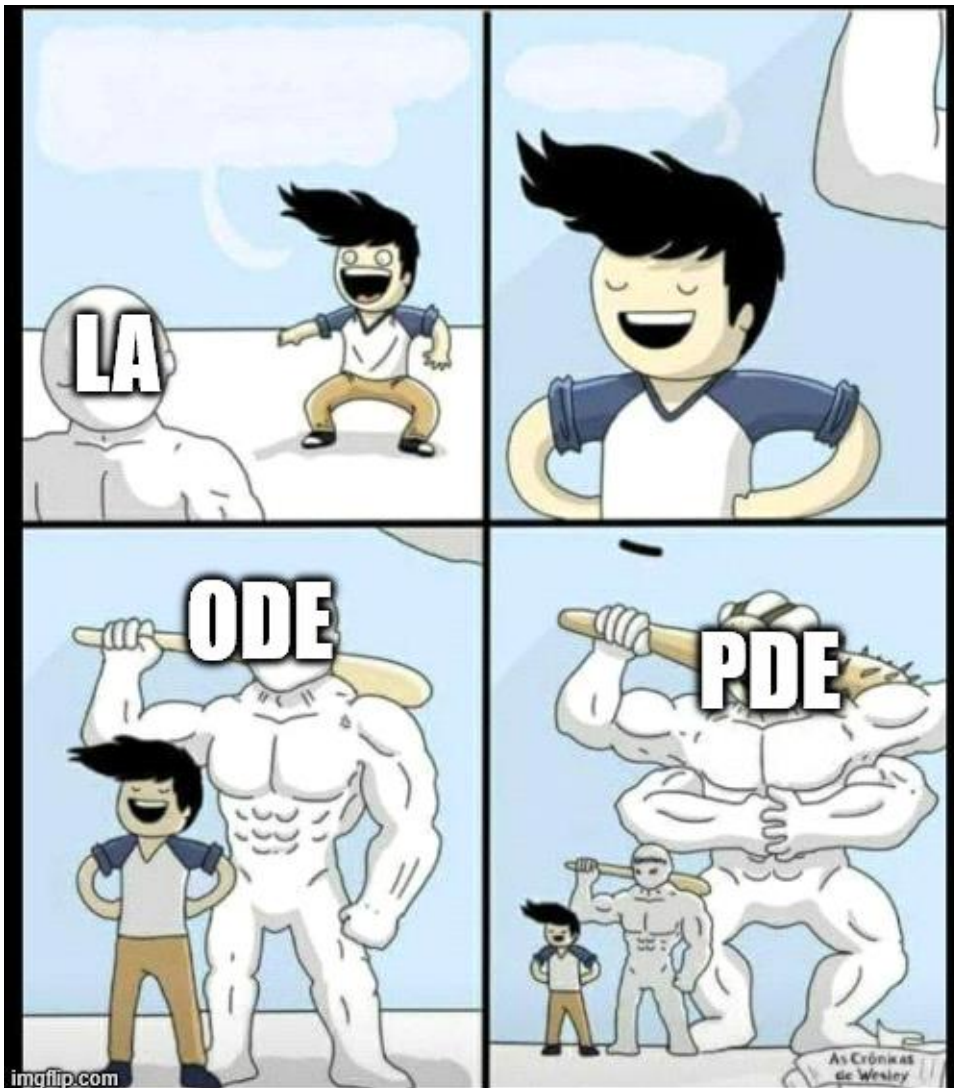


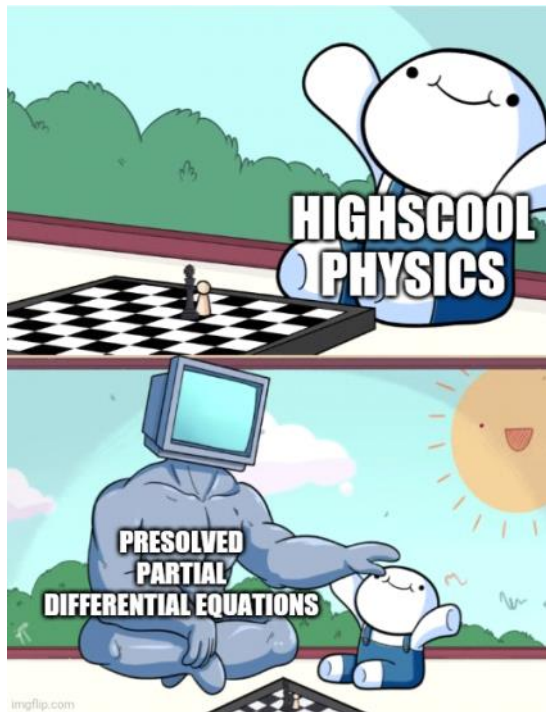


Partially solving a
differential equation



Solving a partial
differential equation





Partial differential equations

In all the above examples we had a function y of a single variable x , y being the solution of an **ordinary** differential equation.

In engineering and science ODEs arise as models for systems where there is **one** independent variable (often x) and **one** dependent variable (often y). Obvious examples are lumped electrical circuits where the current i is a function only of time t (and not of position in the circuit) and lumped mechanical systems (such as the simple harmonic oscillator referred to above) where the displacement of a moving particle depends only on t .

Example 1

Show that $u = \sin x \cosh y$ satisfies the PDE $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.

This PDE is known as **Laplace's equation in two dimensions** and it arises in many applications e.g. electrostatics, fluid flow, heat conduction.

Solution

$$u = \sin x \cosh y \quad \Rightarrow \quad \frac{\partial u}{\partial x} = \cos x \cosh y \quad \text{and} \quad \frac{\partial u}{\partial y} = \sin x \sinh y$$

$$\text{Differentiating again gives } \frac{\partial^2 u}{\partial x^2} = -\sin x \cosh y \quad \text{and} \quad \frac{\partial^2 u}{\partial y^2} = \sin x \cosh y$$

Hence

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -\sin x \cosh y + \sin x \cosh y = 0$$

so the given function $u(x, y)$ is indeed a solution of the PDE.

$$\text{Show that } u = e^{-2\pi^2 t} \sin \pi x \text{ is a solution of the PDE } \frac{\partial^2 u}{\partial x^2} = \frac{1}{2} \frac{\partial u}{\partial t}$$

Answer

$$\frac{\partial u}{\partial t} = -2\pi^2 e^{-2\pi^2 t} \sin \pi x \quad \frac{\partial u}{\partial x} = \pi e^{-2\pi^2 t} \cos \pi x$$

Now find $\frac{\partial^2 u}{\partial x^2}$ and complete the Task:

Your solution

Answer

$$\frac{\partial^2 u}{\partial x^2} = -\pi^2 e^{-2\pi^2 t} \sin \pi x, \text{ and we see that } \frac{\partial^2 u}{\partial x^2} = \frac{1}{2} \frac{\partial u}{\partial t} \text{ as required.}$$

The PDE in the above Task has the general form

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{k} \frac{\partial u}{\partial t}$$

where k is a positive constant. This equation is referred to as the **one-dimensional heat conduction equation** (or sometimes as the **diffusion equation**). In a heat conduction context the dependent variable u represents the temperature $u(x, t)$.

The third important PDE involving two independent variables is known as the **one-dimensional wave equation**. This has the general form

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2}$$

(Note that both partial derivatives in the wave equation are second-order in contrast to the heat conduction equation where the time derivative is first order.)

Example 2

- (a) Verify that $u(x, t) = u_0 \sin\left(\frac{\pi x}{\ell}\right) \cos\left(\frac{\pi ct}{\ell}\right)$ (where u_0 , ℓ and c are constants) satisfies the one-dimensional wave equation.
- (b) Verify the boundary conditions i.e. $u(0, t) = u(\ell, t) = 0$.
- (c) Verify the initial conditions i.e. $\frac{\partial u}{\partial t}(x, 0) = 0$ and $u(x, 0) = u_0 \sin\left(\frac{\pi x}{\ell}\right)$.
- (d) Give a physical interpretation of this problem.

- (a) By straightforward partial differentiation of the given function $u(x, t)$:

$$\frac{\partial u}{\partial x} = u_0 \frac{\pi}{\ell} \cos\left(\frac{\pi x}{\ell}\right) \cos\left(\frac{\pi ct}{\ell}\right) \quad \frac{\partial^2 u}{\partial x^2} = -u_0 \left(\frac{\pi}{\ell}\right)^2 \sin\left(\frac{\pi x}{\ell}\right) \cos\left(\frac{\pi ct}{\ell}\right)$$

$$\frac{\partial u}{\partial t} = -u_0 \left(\frac{\pi c}{\ell}\right) \sin\left(\frac{\pi x}{\ell}\right) \sin\left(\frac{\pi ct}{\ell}\right) \quad \frac{\partial^2 u}{\partial t^2} = -u_0 \left(\frac{\pi c}{\ell}\right)^2 \sin\left(\frac{\pi x}{\ell}\right) \cos\left(\frac{\pi ct}{\ell}\right)$$

We see that $\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2}$ which completes the verification.

- (b) Putting $x = 0$, and leaving t arbitrary, in the given solution for $u(x, t)$ gives

$$u(x, 0) = u_0 \sin 0 \cos\left(\frac{\pi ct}{\ell}\right) = 0 \quad \text{for all } t$$

Similarly putting $x = \ell$, t arbitrary: $u(\ell, 0) = u_0 \sin \pi \cos\left(\frac{\pi ct}{\ell}\right) = 0 \quad \text{for all } t$

- (c) Evaluating $\frac{\partial u}{\partial t}$ firstly for general x and t

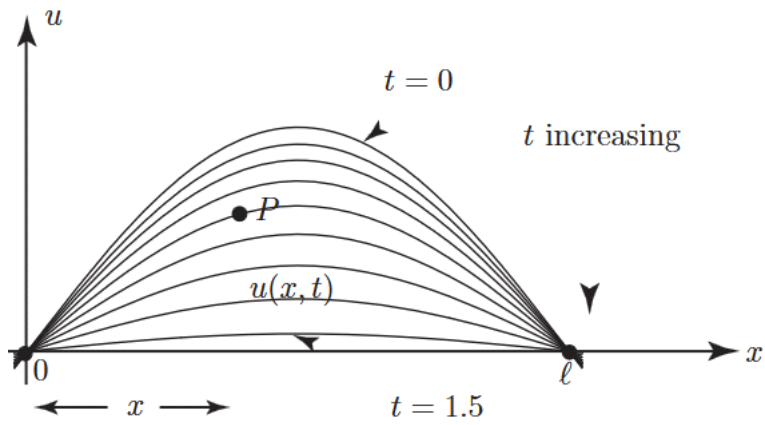
$$\frac{\partial u}{\partial t} = -u_0 \left(\frac{\pi c}{\ell}\right) \sin\left(\frac{\pi x}{\ell}\right) \sin\left(\frac{\pi ct}{\ell}\right)$$

Now putting $t = 0$ leaving x arbitrary

$$\frac{\partial u}{\partial t}(x, 0) = -u_0 \left(\frac{\pi c}{\ell}\right) \sin\left(\frac{\pi x}{\ell}\right) \sin 0 = 0.$$

Also, putting $t = 0$ in the expression for $u(x, t)$ gives

$$u(x, 0) = u_0 \sin\left(\frac{\pi x}{\ell}\right) \cos 0 = u_0 \sin\left(\frac{\pi x}{\ell}\right).$$



Key Point 4

The three PDEs of greatest general interest involving two independent variables are

- (a) The two-dimensional Laplace equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

- (b) The one-dimensional heat conduction equation:

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{k} \frac{\partial u}{\partial t}$$

- (c) The one-dimensional wave equation:

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2}$$